

Transfer Learning versus Training from Scratch for Scene Image Classification

CS 5330 Final Project

Shriman Raghav Srinivasan

Khoury College of Computer Sciences
Northeastern University

July 2026

The Core Question

How much of the benefit of Transfer Learning is due to **pre-trained feature quality**, and how much is due to **reduced trainable capacity** (fewer parameters to optimize)?

- A naive comparison between a pre-trained frozen network and a fully trained random network confounds feature quality and parameter capacity.
- **Our Solution:** A capacity-matched three-condition experimental design:
 - ① **Pretrained-Frozen (PT-F):** Loaded with ImageNet weights; early layers frozen.
 - ② **Random-Frozen (R-F):** Randomly initialized; early layers frozen (Capacity Control).
 - ③ **Random-Full (R-FL):** Randomly initialized; all layers trainable (Scratch Baseline).

Model Architecture & Partitioning

- **Base Model:** ResNet-18 architecture.
- **Frozen Subset (θ_f):** Input convolution (conv1), first batchnorm (bn1), and the first three residual stages (layer1, layer2, layer3) contain 2,782,784 parameters.
- **Trainable Subset (θ_t):** Final residual stage (layer4) and 6-class output head (fc) contain 8,396,806 parameters.
- **Critical Technical Detail:** Batch Normalization layers in θ_f are forced into evaluation mode (`model.eval()`) during training. This prevents running statistics from drifting and ruining the representation.

Intel Image Classification Dataset

- Natural scene images across 6 classes:
Buildings, Forest, Glacier, Mountain, Sea, Street.
- Preprocessing:
 - 1 Resized shorter side to 224px.
 - 2 Center-cropped to 224x224.
 - 3 ImageNet channel mean & std dev normalization.

Splits Structure

- **Training (90%):** 12,631 images.
- **Validation (10%):** 1,403 images (reproducible class-stratified split).
- **Test (Held-out):** 3,000 images (exclusively for final reporting).

Optimizer & Loss

We optimize the trainable parameters θ_t using Stochastic Gradient Descent (SGD) with Cross-Entropy Loss.

- **Hyperparameters:** Learning rate = 0.001, momentum = 0.9, weight decay = 1e-4. Batch size = 32.
- **Early Stopping:** Validation loss is monitored; training stops after 5 consecutive epochs of no improvement.
- **Robustness:** Each of the 3 conditions runs across 3 distinct random seeds (42, 100, 2026) to generate statistics.

Results - Accuracy and Convergence

Table: Test Accuracy and Convergence Speed Comparison

Condition	Mean Test Acc.	Std. Dev.	Mean Epochs
Pretrained-Frozen	93.27%	0.23%	8.0
Random-Frozen	66.54%	4.04%	8.7
Random-Full	84.69%	1.87%	15.0

- **Key Takeaway:** PT-F achieves a massive accuracy boost ($> 50\%$) over R-F, directly isolating the value of ImageNet pretrained representations.

Error Analysis & Class Confusions

- Representative seeds closest to the mean test accuracy:
 - Pretrained-Frozen: Seed 100 (93.23%)
 - Random-Frozen: Seed 100 (66.10%)
 - Random-Full: Seed 100 (84.90%)
- **Common Confusions Across All Conditions:**
 - **Glacier vs. Mountain:** Visually similar snow layers and rock features.
 - **Buildings vs. Street:** Buildings naturally populate street backgrounds.
- Confusion is a property of dataset visual similarity, not the initialization or capacity condition.

- **Pretraining is Essential:** For matched capacities, ImageNet pretraining yields a performance boost of $> 50\%$. Arbitrary random weights in θ_f constrain the representation space.
- **Scratch Limits:** Training from scratch with full capacity (R-FL) outperforms the frozen random control (R-F) but remains far inferior to transfer learning (PT-F), despite taking much longer to converge.
- **Final Conclusion:** In compute-constrained and data-limited settings, pre-trained weights function as critical regularizers and convergence accelerators.